

Distribution System Planning With Reliability

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Abstract—This paper presents a model for solving the multistage planning problem of a distribution network. The objective function to be minimized is the net present value of the investment cost to add, reinforce or replace feeders and substations, losses cost, and operation and maintenance cost. The model considers three levels of load in each node and two investment alternatives for each resource to be added, reinforced or replaced. The nonlinear objective function is approximated by a piecewise linear function, resulting in a mixed integer linear model that is solved using standard mathematical programming. The model allows us to find multiple solutions in addition to the optimal one, helping the decision maker to analyze and choose from a pool of solutions. In addition to the optimization problem, reliability indexes and associated costs are computed for each solution, based on the regulation model used in Brazil. Numerical results and discussion are presented for an illustrative 27-node test network.

Index Terms—Customer interruption cost, economic analysis, power distribution planning, regulation, reliability.

NOMENCLATURE

A) Sets and Indexes

Nd	Number of nodes with existing and new loads.
Ng	Number of available transformers for increasing substation capacity.
Nj	Number of available alternatives to expand the system.
Nk/Nl	Number of available conductors for each branch to be replaced/added.
NLo	Number of load levels.
Nn	Total number of system nodes.
Nq	Number of customer sectors in each node.

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Ns	Number of nodes with existing and new substations.
Nt	Number of <i>in use</i> resources (branches and transformers) at each stage.
B) Parameters	
λ_j	Average failure rate in section j .
η_j	Economic lifetime of available alternative j to expand the system.
μ/ν	Penalty factors settled by regulation.
$CDF(\gamma_j)_q$	Cost of an interruption for an outage duration of γ_j associated with customer type q .
CF_{js}	Fixed investment cost to build a new substation or expand an existing one at node js .
CID_p	Customer interruption duration target settled by regulation.
CIF_p	Customer interruption frequency target settled by regulation.
$C_{ja,l}$	Investment cost of resource l related to the additional branch ja .
$C_{jr,k}$	Investment cost of resource k related to the replacement branch jr .
$CLASS_{d,q}$	Percentage of customer type q in load node d .
$CO_{ja,l}$	Maintenance cost of resource l related to the additional branch ja .
CO_{je}	Maintenance cost of existing branch je that may be replaced.
CO_{jf}	Maintenance cost of fixed branch jf .
$CO_{jr,k}$	Maintenance cost of resource k related to the replacement branch jr .
$CO_{js,g}$	Maintenance cost of transformer g at node js .
CO_{jse}	Maintenance cost of existing transformer at node jse .
$CV_{js,g}$	Investment cost of resource g (transformer) related to a substation at node js .
$D_{d,t,LL}$	Power demand at node d at each load level LL of stage t .

$DURL_{LL}$	Duration in hours of each load level LL .	$Z_{ja,l}$	Impedance of conductor l for the additional branch ja .
$f_{\max}$	Maximum current in cable.	Z_{je}	Impedance of existing branch je that may be replaced.
$F_{ja,l}^{\max(m,n)}$	Maximum branch current in section (m,n) of alternative l in replacement network ja .	Z_{jf}	Impedance of fixed branch jf .
$F_{je}^{\max(m,n)}$	Maximum branch current in section (m,n) of the replacement network je .	$Z_{jr,k}$	Impedance of conductor k to replace branch jr .
$F_{jf}^{\max(m,n)}$	Maximum branch current in section (m,n) of the fixed network jf .	<i>C) Binary Variables</i>	
$F_{jr,k}^{\max(m,n)}$	Maximum branch current in section (m,n) of alternative k of the additional network jr .	$X_{ja,l,t}$	Decision to use conductor l in addition to branch ja at stage t .
I	Interest rate.	$X_{jr,k,t}$	Decision to use conductor k in the replacement branch jr at stage t .
$INV_{\max,j,t}$	Maximum investment cost of alternative j at stage t .	$XF_{js,t}$	Decision to expand a substation at node js at stage t .
$RATE_{LL}$	Cost of energy at each load level LL .	$XV_{js,g,t}$	Decision to use transformer g in the expanding substation capacity at node js at stage t .
$R_{ja,l}$	Resistance of conductor l for the additional branch ja .	$Y_{ja,l,t}$	Utilization of conductor l in the addition branch ja at stage t .
R_{je}	Resistance of existing branch je that may be replaced.	$Y_{je,t}$	Utilization of the existing branch je at stage t .
R_{jf}	Resistance of fixed branch jf .	$Y_{jf,t}$	Utilization of the fixed branch jf at stage t .
$R_{jr,k}$	Resistance of conductor k to replace branch jr .	$Y_{jr,k,t}$	Utilization of the conductor k in replacement branch jr at stage t .
$R_{js,g}$	Resistance of new transformer g at node js .	<i>D) Continuous Variables</i>	
R_{jse}	Resistance of existing transformers located at node jse .	$CIC_{d,j}$	Customer interruption cost of load at node d due to an outage in section j .
$SAIDI_p$	System average interruption duration index target settled by regulation.	$CIDC_t$	Customer interruption duration cost at stage t .
$SAIFI_p$	System average interruption frequency index target settled by regulation.	$CIFC_t$	Customer interruption frequency cost at stage t .
$SE_{\max,js,g}$	Maximum injected current through alternative g of a new transformer at node js at load level LL of stage t .	$F_{jf,t,LL}^{(m,n)}$	Branch current in section (m,n) of the fixed network jf at load level LL of stage t .
$SE_{\max,jse}$	Maximum injected current through an existing transformer at node jse at load level LL of stage t .	$F_{je,t,LL}^{(m,n)}$	Branch current in section (m,n) of the replacement network je at load level LL of stage t .
T	Number of stages of the planning horizon.	$F_{jr,k,t,LL}^{(m,n)}$	Branch current in section (m,n) of alternative k of the additional network jr at load level LL of stage t .
$V_{\max,t,LL}$	Maximum voltage level permitted at each node at load level LL of stage t .	$F_{jse,t,LL}^{(m,n)}$	Branch current in section (m,n) of alternative l in the replacement network ja at load level LL of stage t .
$V_{\min,t,LL}$	Minimal voltage level permitted at each node at load level LL of stage t .	$F_{jse,t,LL}$	Injected current through an existing transformer at node jse at load level LL of stage t .
$VOLL$	Value of lost load (load shedding cost).		

$F_{js,g,t,LL}$	Injected current through alternative g of a new transformer at node js at load level LL of stage t .
$INV_{j,t}$	Investment cost of alternative j at stage t .
$LS_{d,t,LL}$	Load shedding in load node d at load level LL of stage t .
$MAINT_{j,t}$	Maintenance cost of alternative j at stage t .
$OPER_{j,t,LL}$	Operational cost of alternative j at each load level of stage t .
$SAIC_t$	Cost of not attending SAIFI or SAIDI at stage t .
$V_{t,LL}^m$	Voltage at node m at load level LL of stage t .

I. INTRODUCTION

TRADITIONALLY, distribution expansion planning models seek to obtain a minimum cost investment plan along the planning horizon, contemplating the operation of additional, replaced, or reinforced resources to the existing system. The objective of these additions is to meet demand at the minimum possible cost, with acceptable quality and security standards, which are either established by regulation or by customer requirements.

Thus, the problem consists of determining the size, location, and time in which a new resource, such as a feeder or a substation, must be added, replaced, or reinforced [1]–[4].

Given these features and the different aspects considered [3], [4], the problem is eminently nonlinear and multistage, with a significant amount of binary and continuous decision variables, making it a complex problem. Thus, a great variety of models and solution techniques exist [5]–[17]. For instance, [5] uses a modified dynamic programming to solve the expansion problem, but without taking reliability into account.

On the other hand, one of the problems identified in distribution systems planning models is that there is only one optimal plan at minimum cost [3], in which the suboptimal plans are discarded, despite being attractive in terms of adequacy, security, and reliability. Furthermore, there is no comparison of reliability indices before and after the expansion plan, this being one of the key factors for decision makers, either from the regulatory standpoint or from the customers' side. In the latter case, these indices are converted into customer damage functions due to loss produced by service interruptions [18], [19]. The aforementioned reliability indices can guide decision makers, since they can represent filtering mechanisms for plans that deal with specific guidelines of a distribution company (DISCO).

Given the previous context, this paper proposes a novel way to analyze distribution planning expansion, with a multistage model that generates a pool of plans to expand a distribution system with a sequential computation of reliability indices and their associated values or costs. The model takes into account the investment, maintenance, and approximate energy losses

in feeders and substations in the pool generation stage, with the most commonly used distribution system reliability indices computed in the next stage.

This paper is structured as follows. Section II presents the challenges of incorporating reliability to the distribution planning model. Section III depicts the mathematical model formulation, Sections IV and V describe the test system and the results, and Section VI draws conclusions.

II. RELIABILITY IN DISTRIBUTION SYSTEM PLANNING

The majority of the interruptions suffered by customers takes place at the distribution system level, due to its peculiar characteristics and structure [20], [21], the reason for which there is a growing interest in distribution expansion planning models that incorporate reliability issues.

It is essential for utilities to know the reliability level of their existing systems and to foresee the evolution of the most important performance indices before and after expansions. The objective is to check whether all customers will get served in a reliable way and at an acceptable cost, this being defined either by the regulator on behalf of the customers, or by the customers themselves through customer damage functions.

Therefore, it is necessary to determine the long-term behavior of the system considering: 1) failure rates of system components, which are known for existing components and estimated for future additions along the time horizon and 2) duration of the interruptions as a function of the repair time, service recovery, switching, or isolation states.

To directly evaluate the value of reliability is a difficult task, usually associated with customer interruption costs and represented through a damage function for each customer type, which is obtained through surveys [19], [22], [23]. The value that the interruptions have for the consumer is the estimated cost that he or she is willing to pay to receive redundancy in the system in order to avoid interruption. This value can be used to create adequate signals to guide investments in the distribution network by establishing an upper bound for the customer interruption cost and, if the cost exceeds this upper bound, then the allowed revenue for the distribution company is reduced.

The problem of using customer damage functions is that customers' preferences change with time. The same occurs with the continuity of service levels and the costs of the distribution companies. On the other hand, the regulators' task is to serve their customers, ensuring them reliable service, and establishing minimum reliability levels based on historical data in specific locations and times. In general, these are rules of thumb based on the perception of the customers, which are easy to implement and interpret. The levels that are considered acceptable for the continuity of service from the regulator's standpoint should be based on the explicit knowledge of the perception of the customers' tolerance levels and the cost of the losses suffered by them [24], [25]. This is a very complex task for the regulator standpoint.

The two previous ways of considering the cost associated with reliability as an incentive to distribution system expansion can be found in regulations of various countries, as in the cases of Norway and Sweden [26], [27].

The difficulty to incorporate reliability into distribution expansion models resides in the fact that it is necessary to know the

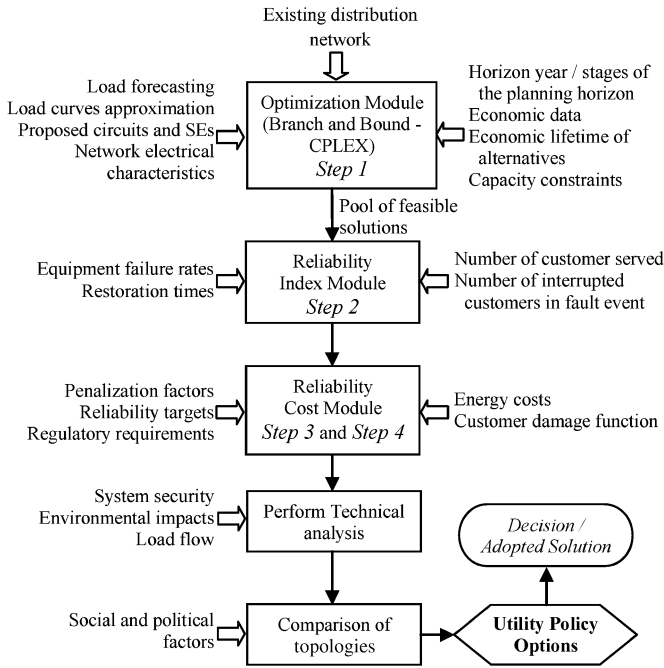


Fig. 1. Distribution planning algorithm flowchart.

network topology in order to calculate the reliability indices that characterize the system as well as the failure and repair rates and the placement and response of the protective devices [28]. However, the final optimal topology is the objective of the expansion plan, where reliability indices are known after expansion in a sequential process [29]. As done with [30], we calculate the costs related to reliability for each loading condition, given previous knowledge of the network. This latter work also proposed some investment alternatives for the existing network, and an exhaustive simulation for each of them is performed to determine the one that gives minimum investment and reliability costs.

With the previous problem characterization, the planning process of distribution systems with reliability is summarized in Fig. 1. The scope of this work is limited to the following steps.

- Step 1) Determine the minimal cost expansion plan and obtain a pool of solutions with a diversity of plans.
- Step 2) Calculate the reliability indices of each distribution system configuration in the pool of solutions for each year of the planned horizon.
- Step 3) Calculate the annual cost of each distribution system configuration in the pool of solutions for the DISCO to obtain the reliability index targets fixed for a specific regulation.
- Step 4) Calculate the annual customer interruption cost (CIC) of each distribution system configuration in the pool of solutions.

Reliability indices and their associated costs are just used as indicators to provide a broad solution scope. Radiality in each stage of the planning horizon is guaranteed by forcing the program to select only one input (feeding) path for each load node. So the presented algorithm is suitable for optimization of radial networks with a single voltage level.

Regarding the inclusion of automated devices, they could be added after adding the branches and substations first, for example.

It is known that the planning process within a distribution utility is conducted heuristically through expert judgment and empirical analysis, with several objectives and constraints that are difficult to formulate efficiently or cannot be expressed in a conventional optimization problem. So an expert planner can derive a small number of possible solution alternatives where new iterations occur as the plans circulate among different departments. In such cases, obtaining multiple solutions will help the expert to choose a small number of plans which best fits all or most of the criteria that could not be expressed in a conventional way.

Note that although we have restricted our analysis to the four steps described in Fig. 1, a technical analysis should be advisable to evaluate the environmental impacts and security issues related to the solution alternatives. These issues, combined with social and political factors, should be discretionarily used by the planners and decision makers, along with the company operation policy, in the comparison of topologies that have exceeded the stage of technical analysis. This would raise a set of policy options from which to select the final solution to be adopted.

III. PROBLEM FORMULATION

The following formulation is performed from a DISCO's perspective, whose obligation is to attend its customers respecting the established reliability limits. The DISCO expands the network using lines and/or substations so that energy supply for current and future customers is served at minimum cost and with the quality required either by customers or regulators. Thus, the impact of reliability on the expansion cost of a DISCO is significant.

The implemented model considers that there are several alternatives for each expansion asset available, for example, depending on the size of the conductors or the transformer's capacity. The model is a multiperiod one, in which each period can have several load levels, describing a typical daily load curve with maximum consumption occurring in each node at different times [7]. The load is represented as a constant current so that the planning model becomes a mixed-integer linear programming problem (MILP). Quadratic losses in lines and transformers are transformed by piecewise linearization so that the problem becomes a mixed-integer linear one, solved by using GAMS/CPLEX [31]. The utilization of MILP techniques ensures that personal computers can achieve fast and efficient solutions for large-size problems.

A. Objective Function and Constraints

The objective function represents the net present value of the cost of expansion (TPV), which is composed of four components: the investment present value (IPV), the maintenance present value (MPV), the operational present value (OPV), and the load shedding present value (LSPV)

$$\min \text{TPV} = \sum_{t=1}^T (\text{IPV} + \text{MPV} + \text{OPV} + \text{LSPV}) \quad (1)$$

where

$$IPV = \sum_{j=1}^{Nj} \left[\frac{(1+I)^{n_j}}{(1+I)^{n_j} - 1} (1+I)^{-t} \right] INV_{j,t} \quad (2)$$

$$INV_{j,t} = \sum_{k=1}^{Nk} C_{jr,k} X_{jr,k,t} + \sum_{l=1}^{Nl} C_{ja,l} X_{ja,l,t} + CF_{js} X_{F_{js,t}} + \sum_{g=1}^{Ng} CV_{js,g} XV_{js,g,t} \quad (3)$$

$$MPV = \sum_{j=1}^{Nt} \left[(1+I)^{-t} + \frac{(1+I)^{-T}}{I} \right] \times MAINT_{j,t} \quad (4)$$

$$MAINT_{j,t} = CO_{jf} Y_{jf,t} + CO_{je} Y_{je,t} + \sum_{k=1}^{Nk} CO_{jr,k} Y_{jr,k,t} + \sum_{l=1}^{Nl} CO_{ja,l} Y_{ja,l,t} + CO_{jse} + \sum_{g=1}^{Ng} CO_{js,g} XV_{js,g,t} \quad (5)$$

$$OPV = \left\{ \left[(1+I)^{-t} + \frac{(1+I)^{-T}}{I} \right] \times \sum_{LL=1}^{NLo} \left(8760 * RATE_{LL} \frac{DURL_{LL}}{24} \right) \right\} * \sum_{j=1}^{Nt} OPER_{j,t,LL} \quad (6)$$

$$OPER_{j,t,LL} = R_{jf} \left(F_{jf,t,LL}^{(m,n)} + F_{jf,t,LL}^{(n,m)} \right)^2 + R_{je} \left(F_{je,t,LL}^{(m,n)} + F_{je,t,LL}^{(n,m)} \right)^2 + \sum_{k=1}^{Nk} R_{jr,k} \left(F_{jr,k,t,LL}^{(m,n)} + F_{jr,k,t,LL}^{(n,m)} \right)^2 + \sum_{l=1}^{Nl} R_{ja,l} \left(F_{ja,l,t,LL}^{(m,n)} + F_{ja,l,t,LL}^{(n,m)} \right)^2 + R_{jse} (F_{jse,t,LL})^2 + \sum_{g=1}^{Ng} R_{js,g} (F_{js,g,t,LL})^2. \quad (7)$$

The terms in (7) are quadratic and, in this paper, we proceed to linearize them in a piecewise fashion, as done in [1] and [10]. The details on how the next equation can be linearized can be found there.

The cost of load shedding is very high compared with the cost of building new lines or substations. This value can be thought of as the value of lost load (VOLL)

$$LSPV = \sum_{d=1}^{Nd} \sum_{LL=1}^{NLo} \left[(1+I)^{-t} + \frac{(1+I)^{-T}}{I} \right] 8760 * VOLL * LS_{d,t,LL}. \quad (8)$$

The constraints considered in the problem are shown to follow [6], [9], [10]:

$$\sum_{n=1}^{Nn} \left\{ \left(F_{jf,t,LL}^{(m,n)} - F_{jf,t,LL}^{(n,m)} \right) + \left(F_{je,t,LL}^{(m,n)} - F_{je,t,LL}^{(n,m)} \right) + \sum_{k=1}^{Nk} \left(F_{jr,k,t,LL}^{(m,n)} - F_{jr,k,t,LL}^{(n,m)} \right) + \sum_{l=1}^{Nl} \left(F_{ja,l,t,LL}^{(m,n)} - F_{ja,l,t,LL}^{(n,m)} \right) \right\} = SE_{s,t,LL} - D_{d,t,LL} + LS_{d,t,LL} \quad \forall (m/n \neq n, t, LL) \quad (9)$$

where

$$SE_{s,t,LL} = F_{jse,t,LL} + \sum_{g=1}^{Ng} F_{js,g,t,LL} \quad s = 1, \dots, Ns \quad d = 1, \dots, Nd \quad Ns + Nd = Nn$$

$$\begin{cases} Z_{jf} \left(F_{jf,t,LL}^{(m,n)} - F_{jf,t,LL}^{(n,m)} \right) = V_{t,LL}^m - V_{t,LL}^n & \forall (m, n/m \neq n, t, LL) \\ Z_{je} \left(F_{je,t,LL}^{(m,n)} - F_{je,t,LL}^{(n,m)} \right) = V_{t,LL}^m - V_{t,LL}^n & \forall (m, n/m \neq n, t, LL) \\ Z_{jr,k} \left(F_{jr,k,t,LL}^{(m,n)} - F_{jr,k,t,LL}^{(n,m)} \right) = V_{t,LL}^m - V_{t,LL}^n & \forall (m, n/m \neq n, k, t, LL) \\ Z_{ja,l} \left(F_{ja,l,t,LL}^{(m,n)} - F_{ja,l,t,LL}^{(n,m)} \right) = V_{t,LL}^m - V_{t,LL}^n & \forall (m, n/m \neq n, l, t, LL). \end{cases} \quad (10)$$

Each of the previous constraints is only active when the respective line is used; thus, each constraint must be multiplied by its respective usage variable, introducing nonlinearities into the problem, which can be avoided by using a disjunctive model as in [6]

$$\begin{cases} 0 \leq F_{jf,t,LL}^{(m,n)} \leq Y_{jf,t} F_{max_{jf}}^{(m,n)} & \forall (m, n/m \neq n) \\ 0 \leq F_{je,t,LL}^{(m,n)} \leq Y_{je,t} F_{max_{je}}^{(m,n)} & \forall (m, n/m \neq n) \\ 0 \leq F_{jr,k,t,LL}^{(m,n)} \leq Y_{jr,k,t} F_{max_{jr,k}}^{(m,n)} & \forall (m, n/m \neq n, k) \\ 0 \leq F_{ja,l,t,LL}^{(m,n)} \leq Y_{ja,l,t} F_{max_{ja,l}}^{(m,n)} & \forall (m, n/m \neq n, l) \end{cases} \quad (11)$$

$$0 \leq SE_{s,t,LL} \leq SE_{max_{jse}} + \sum_{\tau=1}^t XV_{js,g,\tau} SE_{max_{js,g}} \quad \forall g \quad (12)$$

$$\begin{cases} \sum_{t=1}^T \sum_{k=1}^{Nk} X_{jr,k,t} \leq 1 \\ \sum_{t=1}^T \sum_{l=1}^{Nl} X_{ja,l,t} \leq 1 \end{cases} \quad (13)$$

$$\begin{cases} \sum_{t=1}^T XF_{js,t} \leq 1 \\ \sum_{t=1}^T \sum_{g=1}^{Ng} XV_{js,g,t} \leq 1 \end{cases} \quad (14)$$

$$XV_{js,g,t} \leq \sum_{\tau=1}^t XF_{js,\tau} \quad \forall t = 1, \dots, T \quad (15)$$

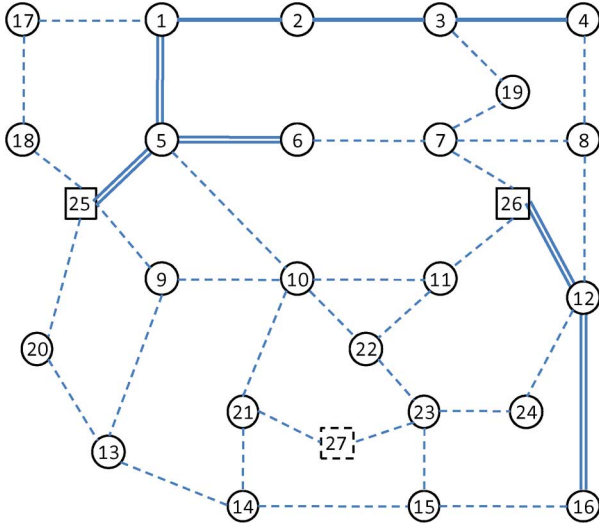


Fig. 2. Diagram of the 27-node network.

 TABLE I
NODAL DATA OF THE NETWORK

Node	Load [MVA]								
	Stage 1			Stage 2			Stage 3		
	LL1	LL2	LL3	LL1	LL2	LL3	LL1	LL2	LL3
1	1.2	0.72	0.24	1.2	0.72	0.24	1.2	0.72	0.24
2	0	0	0	1.2	0.72	0.24	1.2	0.72	0.24
3	0	0	0	0	0	0	1.2	0.72	0.24
4	1.2	0.72	0.24	1.2	0.72	0.24	1.2	0.72	0.24
5	1.2	0.72	0.24	1.2	0.72	0.24	1.2	0.72	0.24
6	1.2	0.72	0.24	1.2	0.72	0.24	1.2	0.72	0.24
7	0	0	0	1.2	0.72	0.24	1.2	0.72	0.24
8	1.2	0.72	0.24	1.2	0.72	0.24	1.2	0.72	0.24
9	0	0	0	1.2	0.72	0.24	2.4	1.2	0.48
10	0	0	0	0	0	0	2.4	3.6	1.2
11	1.2	0.72	0.24	1.2	2.4	0.48	2.4	3.6	1.2
12	0	0	0	1.2	0.72	0.24	1.2	0.72	0.24
13	1.2	0.72	0.24	1.2	2.4	0.48	2.4	3.6	1.2
14	0	0	0	0	0	0	2.4	1.2	0.48
15	0	0	0	0	0	0	2.4	1.2	0.48
16	1.2	0.72	0.24	1.2	0.72	0.24	1.2	0.72	0.24
17	0	0	0	1.2	0.72	0.24	2.4	1.2	0.48
18	0	0	0	0	0	0	1.2	2.4	0.48
19	1.2	0.72	0.24	1.2	2.4	0.48	1.2	2.4	0.48
20	0	0	0	1.2	0.72	0.24	1.2	0.72	0.24
21	1.2	0.72	0.24	1.2	2.4	0.48	1.2	2.4	0.48
22	0	0	0	0	0	0	1.2	0.72	0.24
23	0	0	0	0	0	0	1.2	0.72	0.24
24	1.2	0.72	0.24	1.2	0.72	0.24	1.2	0.72	0.24

 TABLE II
BRANCH DATA OF THE NETWORK

Existing Lines			Conductor Option 1				Conductor Option 2			
$f_{\max}$ [pu]	Z [Ω]	R [Ω]	$f_{\max}$ [pu]	Z [Ω]	R [Ω]	C [k\$]	$f_{\max}$ [pu]	Z [Ω]	R [Ω]	C [k\$]
<i>Installed network</i>										
6	1	0.76	---	---	---	---	---	---	---	---
<i>Replacement network</i>										
6	1	0.76	9.6	0.7	0.44	20	12	0.5	0.25	38
<i>Addition network</i>										
---	---	---	6	1	0.76	90	9.6	0.7	0.44	110

 TABLE III
TRANSFORMERS DATA OF THE NETWORK

Node	Existing			CF [k\$]	Transformer Option 1				Transformer Option 2			
	$f_{\max}$ [pu]	R [Ω]	CO [k\$]		$f_{\max}$ [pu]	R [Ω]	CO [k\$]	CV [\$]	$f_{\max}$ [pu]	R [Ω]	CO [k\$]	CV [\$]
25	15	0.13	3	50	7.5	0.25	1	375	12	0.16	2	600
26	15	0.13	3	70	7.5	0.25	1	375	12	0.16	2	600
27	---	---	---	70	7.5	0.25	1	375	12	0.16	2	600

 TABLE IV
NUMBER OF CONSUMERS PER NODE AND SECTOR PARTICIPATION IN LOAD

Node	Stage 1				Stage 2				Stage 3			
	NC	Res	Com	Ind	NC	Res	Com	Ind	NC	Res	Com	Ind
1	100	0.5	0.2	0.3	100	0.5	0.2	0.3	100	0.5	0.2	0.3
2	---	---	---	---	10	0	0.1	0.9	10	0	0.1	0.9
3	---	---	---	---	---	---	---	---	50	0.2	0.2	0.6
4	10	0	0.2	0.8	10	0	0.2	0.8	10	0	0.2	0.8
5	50	0.2	0.2	0.6	50	0.2	0.2	0.6	50	0.2	0.2	0.6
6	30	0.15	0.2	0.65	30	0.15	0.2	0.65	30	0.15	0.2	0.65
7	---	---	---	---	10	0	0	1	10	0	0	1
8	50	0.1	0.2	0.7	50	0.1	0.2	0.7	50	0.1	0.2	0.7
9	---	---	---	---	50	0.1	0.2	0.7	100	0.1	0.2	0.7
10	---	---	---	---	---	---	---	---	10	0	0	1
11	50	0.5	0.4	0.1	50	0.5	0.4	0.1	70	0.5	0.4	0.1
12	---	---	---	---	50	0.4	0.4	0.2	50	0.4	0.4	0.2
13	10	0	0.15	0.85	10	0	0.15	0.85	15	0	0.15	0.85
14	---	---	---	---	---	---	---	---	15	1	0	0
15	---	---	---	---	---	---	---	---	15	1	0	0
16	50	0.3	0.2	0.5	50	0.3	0.2	0.5	50	0.3	0.2	0.5
17	---	---	---	---	10	0	0.25	0.75	15	0	0.25	0.75
18	---	---	---	---	---	---	---	---	20	0.8	0.2	0
19	10	0	0	1	10	0	0	1	10	0	0	1
20	---	---	---	---	50	0	0	1	50	0	0	1
21	100	0.6	0.25	0.15	100	0.6	0.25	0.15	100	0.6	0.25	0.15
22	---	---	---	---	---	---	---	---	10	0	1	0
23	---	---	---	---	---	---	---	---	10	1	0	0
24	50	0.5	0.2	0.3	50	0.5	0.2	0.3	50	0.5	0.2	0.3

$$\begin{cases} Y_{jf,t} \leq 1 \\ Y_{jr,k,t} \leq \sum_{\tau=1}^t X_{jr,k,\tau} \\ Y_{ja,l,t} \leq \sum_{\tau=1}^t X_{ja,l,\tau} \end{cases} \quad (16)$$

$$Y_{je,t} \leq 1 - \sum_{\tau=1}^t X_{jr,k,\tau} \quad (17)$$

$$\begin{aligned} & \sum_{k=1}^{Nk} C_{jr,k} X_{jr,k,t} + \sum_{l=1}^{Nl} C_{ja,l} X_{ja,l,t} + CF_{js} X F_{js,t} \\ & + \sum_{g=1}^{Ng} CV_{js,g} X V_{js,g,t} \leq INV_{\max t} \end{aligned} \quad (18)$$

$$V_{\min t,LL} \leq V_{t,LL}^m \leq V_{\max t,LL} \quad (19)$$

$$0 \leq LS_{d,t,LL} \leq D_{d,t,LL} \quad (20)$$

B. Reliability Indices Computation

After the solution pool is obtained, reliability indices are computed. The most commonly used indices in distribution system reliability are system average interruption frequency index (SAIFI), system average interruption duration index (SAIDI), average system availability index (ASAI), and customer interruption duration (CID) at each load node, Customer

interruption frequency (CIF) at each load node and expected energy not supply (EENS) [20]–[25], [28], [29].

The calculation of the previous indices considers that each feeder has a circuit breaker without a recloser at the output of the substation and that each section between nodes has a switch that enables the reconfiguration of the system after a fault in order to meet the loads as much as possible [28], [32].

C. Reliability Cost Computation

The cost of nonsupplied energy can be considered either from the DISCOs' viewpoint or from the customers' viewpoint. From the DISCOs' standpoint, this cost corresponds to the energy that is not billed during the fault, and from the customers' view, is related to each customer's damage function or interruption cost function, being both functions that are difficult to measure. This paper assumes that the regulator sets limits for the reliability indices and penalizes its noncompliance. Thus, DISCOs expand the system by seeking the minimum cost and observing these indices and their associated penalties for nonfulfillment.

From this perspective, the EENS cost at each stage is

$$\text{EENSC}_t = \sum_{d=1}^{Nd} \sum_{LL=1}^{NLo} \left(\frac{\text{DURL}_{LL}}{24} \right) \times D_{d,t,LL} \text{CID}_{d,t} \text{RATE}_{LL}. \quad (21)$$

The way to penalize the nonattendance of the reliability indices depends on the specific regulation. This work uses the Brazilian regulation, which specifies that the DISCOs must remunerate the customer, whose reliability index is violated according to its energy bill during the considered period, and with a penalty factor μ established by the regulator.

Therefore, the CID and CIF costs at each stage are

$$\begin{aligned} \text{CIDC}_t &= \mu \sum_{d=1}^{Nd} \left[(\text{CID}_{d,t} - \text{CID}_p) \sum_{LL=1}^{NLo} \left(\frac{\text{DURL}_{LL}}{24} \right) \right. \\ &\quad \times D_{d,t,LL} \text{RATE}_{LL}] \\ &\quad \text{if } (\text{CID}_{d,t} > \text{CID}_p) \\ \text{CIFC}_t &= \mu \sum_{d=1}^{Nd} \left[(\text{CIF}_{d,t} - \text{CIF}_p) \sum_{LL=1}^{NLo} \left(\frac{\text{DURL}_{LL}}{24} \right) \right. \\ &\quad \times D_{d,t,LL} \text{RATE}_{LL}] \\ &\quad \text{if } (\text{CIF}_{d,t} > \text{CIF}_p). \end{aligned} \quad (22)$$

The highest of the previous values is passed on to the affected customers in proportion to their individual bills.

The economic valuation of not attending the average frequency values or the duration of faults in a considered period also depends on the specific regulation. In this paper, we consider the penalty applied by the regulator as a percentage ν of the energy bill of the DISCO's sold energy.

Thus, the SAIFI or SAIDI cost at each stage is

$$\text{SAIC}_t = \nu \sum_{d=1}^{Nd} \sum_{LL=1}^{NLo} \left(\frac{\text{DURL}_{LL}}{24} \right) D_{d,t,LL} 8760 * \text{RATE}_{LL} \text{ if } (\text{SAIFI}_t > \text{SAIFI}_p) \text{ or } (\text{SAIDI}_t > \text{SAIDI}_p). \quad (23)$$

The cost for not complying with the CID or the CIF, which is directly passed on to the affected customers, must be discounted

from the value of the penalty for not satisfying the SAIFI or SAIDI.

The cost associated with the customers' reliability can be obtained from cost damage functions [24], [32].

The customer interruption cost (CIC) of the load at node d , due to an outage in section j in year t is

$$\begin{aligned} \text{CIC}_{d,j} &= \lambda_j \sum_{q=1}^{Nq} \text{CLASS}_{d,q} \text{CDF}(\gamma_j)_q \\ &\quad \times \sum_{LL=1}^{NLo} \left(\frac{\text{DURL}_{LL}}{24} \right) D_{d,t,LL}. \end{aligned} \quad (24)$$

The total cost of interruptions for a load at node d can be determined by adding the cost of all section outages. The total cost of customer interruptions for all customers can then be evaluated [32]. The CIC represents the cost that the customer experiences and provides a means to reduce the allowed revenue for the DISCO if that is above a maximum established by the regulator.

IV. TEST DISTRIBUTION NETWORK DESCRIPTION

With the objective of showing the results obtained with the model proposed, we have used a fictitious system composed of 24 nodes, 3 substation nodes, and 39 branches operating at 13.8 kV, as adapted from [7]. The topology of the system is shown in Fig. 2, where solid line squares denote existing substations that can be reinforced, and dashed squares represent previously nonexistent candidate substations. Circles represent load nodes, considered to be concentrated in this model, whose data are shown in Table I. Load levels $LL1$, $LL2$, and $LL3$ have a duration of 3, 13, and 8 h/day, respectively.

The solid lines depict branches that are part of the existing network and cannot be reinforced or replaced, and double solid lines depict branches that can be replaced. Branches that can be added to the current network are illustrated by dashed lines. Five of the existing lines can be replaced with two possible conductor sizes. For the 31 lines that can be added, there are two possible conductor sizes. Line data are shown in Table II. Substation data are presented in Table III. Base values are 13.8 kV and 1 MVA.

This work has adopted a planning horizon of 3 years subdivided into periods of 1 year, assuming that, from year T onwards, maintenance, operation, and reliability costs remain constant, as in [2].

The capital recovery rate is set at 10%, and the line and transformer lifetimes are 25 and 15 years, respectively.

Electrical tariff evaluates losses in lines and transformers, and the penalties for not reaching the required reliability indices. Their value is set at U.S.\$50/MWh, U.S.\$40/MWh, and U.S.\$27.4/MWh for load levels $LL1$, $LL2$, and $LL3$, respectively. For simplicity, maintenance cost is set at U.S.\$450/yr. To evaluate possible load shedding due to a lack of available capacity, the VOLL is set at U.S.\$2000/MWh. Upper and lower voltage limits are 0.95 and 1.05 p.u., respectively, where the voltage at the node with a fixed substation is equal to 1.05 p.u.

The quadratic variable cost of energy losses in feeders and substations is approximated by three linear segments.

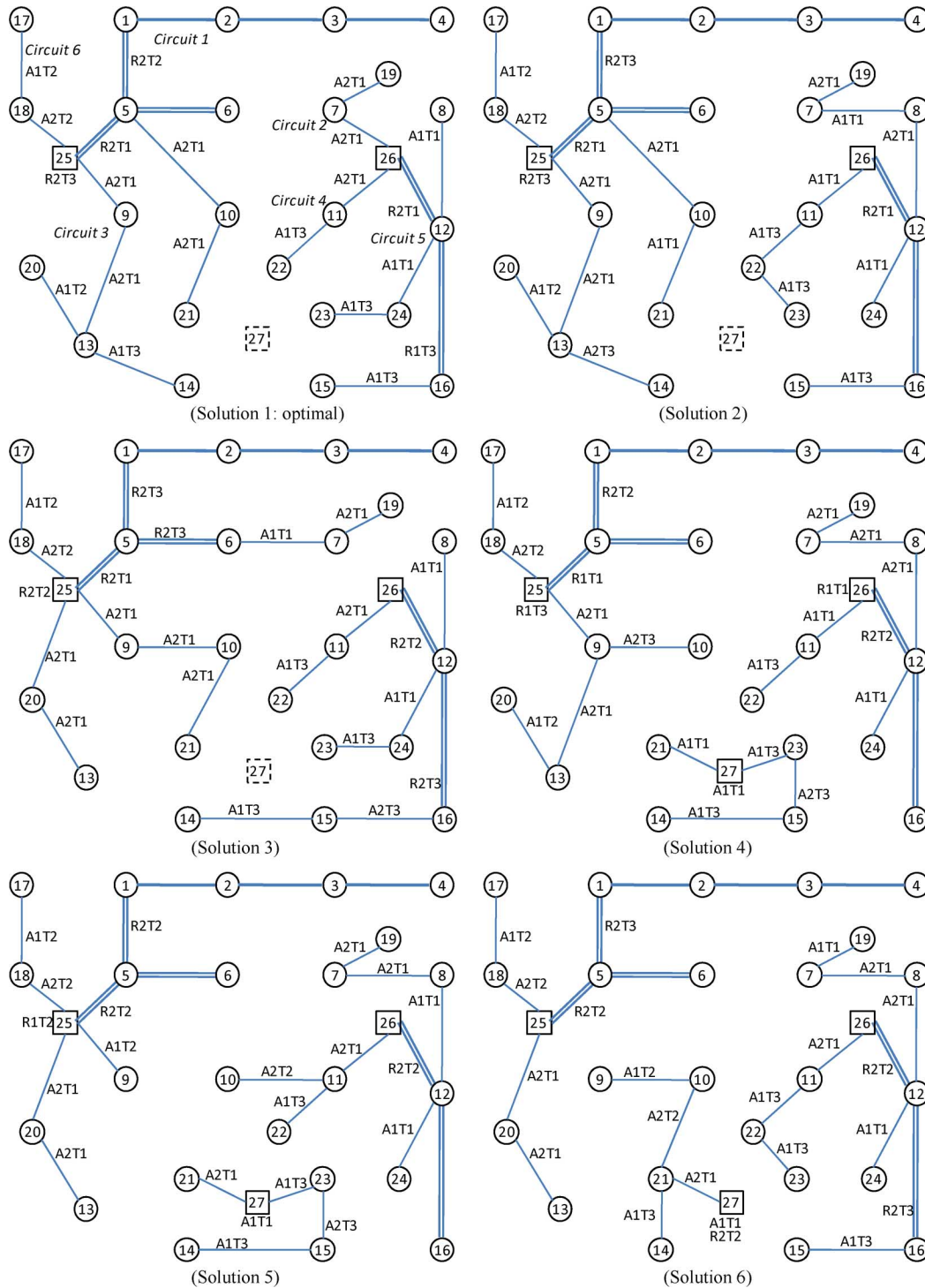


Fig. 3. Pool of solutions of the 27-node network.

For the sake of simplicity, a failure rate of 0.4 failures per year, repair times of 2 h, and reconfiguration times of 0.25 h, are assumed for all lines. The penalty factor μ for not meeting the CID or CIF is set at 20, and the ν applied as a penalty to the annual electricity billing statement of a distribution company (DISCO) is set at 1%.

Reliability indices must meet the limits imposed by the regulator in order for the DISCO to not be penalized. In this paper, we have adopted the following values: $CID_p = 4$ h/yr, $CIF_p = 4$

interruptions per year, $SAIDI_p = 2$ h/yr, and $SAIFI_p = 2$ interruptions per year. The number of customers and its distribution as residential, commercial, or industrial are shown in Table IV. The values of the interruption costs per customer sector and interruption durations can be seen in Table V.

V. NUMERICAL RESULTS

With the objective of validating the model developed, several simulations were carried out using the CPLEX solver in GAMS

TABLE V
SECTOR INTERRUPTION COST

Customer sector	Interruption Cost [\$/kW]	
	15 min.	120 min.
Residential	0.1	5
Commercial	3	32
Industrial	4	26

[31]. The results presented correspond to the expansion of the system in which investments are restricted to a maximum of U.S. k\$2000/expansion period.

CPLEX provides a procedure to specifically generate multiple solutions. There are two ways to fill the solution pool associated with the model: accumulating successive incumbents or generating alternative solutions by populating the solution pool. It is possible to invoke the procedure many times in a row in order to explore the solution space differently. In particular, you may invoke this procedure multiple times to find additional solutions, especially if the first solutions found are not satisfactory. This is done by specifying a GAMS program that inspects the solutions.

Note that other works, such as [33], propose construction plans by using heuristic techniques that do not guarantee obtaining an optimal solution but generate feasible, good-quality solutions.

In this paper, the feasible solutions chosen to be part of the pool are those whose objective function values are, at most, 10% higher than the optimal solution, but other gap values can be selected.

A filter is applied to select solutions whose configurations are different in at least one branch, the solutions being classified according to the value of the objective function. Other filters can be used to select solutions whose settings are different in at least two branches, or include the addition of a new substation or transformer.

The values in Table VI are the costs corresponding to the expansion and reliability of the network for some solutions in the pool. The net present value of investment, maintenance, and operation costs (TPV) for each resulting expansion plan is shown in the first line of Table VI, and the other costs are presented for each expansion period. This way, the decision maker can have more information regarding each plan.

Fig. 3 shows the network topology for the same solutions in Table VI, investment constraints included, where the first network depicts the global optimal solution (solution 1). In this figure, letter *A* corresponds to additional lines, letter *R* corresponds to replaced lines, and the number following these letters is the option number, either 1 or 2, related to the size of the conductor and, finally, letter *T* and the following number are related to the period in which the asset is added or replaced (1, 2, or 3).

The results show the diversity of solutions, from the costs distribution across time and resulting topologies standpoints, without limiting the decision maker to a single plan that many times does not meet other specific political, social, and strategic constraints. Another thing that can be observed from the topologies in Fig. 3 is that particular lines are always present, as is the case of the lines between nodes 18 and 25, 17 and 18, 11 and

TABLE VI
COSTS OF THE POOL OF SOLUTIONS IN (K)

COST	Solution Number					
	1	2	3	4	5	6
TPV	6107.36	6238.28	6329	6554.39	6620.89	6635.95
<i>Stage 1</i>						
INV	1726	1482	1806	1988	2000	2000
MAINT	13.65	13.65	14.1	14.2	14.2	14.2
OPER	37.37	40.23	45.23	48.05	53.66	54.44
EENSC	0.569	0.630	0.602	0.557	0.557	0.557
CIDC	0.506	0.506	0.506	0.506	0.506	0.506
SAIC	24.387	24.387	24.387	24.387	24.387	24.387
CIC	270.33	297.596	296.193	280.905	280.905	280.905
<i>Stage 2</i>						
INV	550	512	1097	587	1375	1295
MAINT	15	15	17	15.55	17	18
OPER	87.45	118.05	115.38	102.91	91.85	96.85
EENSC	1.092	1.273	1.186	1.114	1.071	1.118
CIDC	0.506	0.506	0.506	0.506	0.506	0.506
SAIC	51.197	51.197	51.197	51.197	51.197	51.197
CIC	548.094	643.878	616.252	602.403	576.082	601.169
<i>Stage 3</i>						
INV	1092	1140	559	1187	642	512
MAINT	18.8	18.8	18.8	18.8	18.8	19.8
OPER	246.87	286.82	249.65	244.22	217.51	234.01
EENSC	2.015	2.205	2.098	1.877	1.804	1.988
CIDC	0.506	0.506	0.506	0.506	0.506	0.506
SAIC	91.051	91.051	91.051	91.051	91.051	91.051
CIC	885.57	982.246	951.476	924.594	895.189	948.397

26, or 11 and 22, suggesting that these are lines that are part of a plan that can be considered robust.

As shown in Table VI, the models allow us to know the cost associated with the reliability indices more commonly used in distribution systems. In our example, the results are presented per period to show that costs, updated by their present value, evolve along the planning horizon, which is a relevant feature for a decision maker.

One thing that must be highlighted in the values obtained is that the cost incurred by the DISCO for not meeting the established reliability limits settled by the regulator can be significantly lower than the customer interruption cost (CIC). Thus, the CIC can be used to give incentives which represent a good balance between high redundancy and cost effectiveness corresponding to the estimated cost that the customers are willing to pay for to receive redundancy.

Other relevant information added to the reliability costs is the time evolution of the indices along the planning horizon per node or per circuit. Table VII shows the CID, CIF, and EENS at each node of the network for the optimal solution, showing the evolution of these indices. The observation of these values shows that the regulatory CIF limit is never surpassed at any node or period, but the CID limit is surpassed at node 4 during all 3 periods. Many strategies to introduce corrective measures, such as the addition of circuit breakers or circuit interconnections, can be visualized in different topologies, similar to the ones shown in Fig. 3.

As presented in Table VIII, the reliability indices per customer group are also calculated per circuit leaving the substation (see Solution 1 of Fig. 3). The values indicate that SAIDI and SAIFI surpass the limits imposed by regulation in circuit

TABLE VII
RELIABILITY INDICES AT SOME NODES FOR THE OPTIMAL SOLUTION

Node	CIF [interruptions]			CID [hours]			EENS [MWh]		
	Stage								
	1	2	3	1	2	3	1	2	3
1	3.2	3.2	3.2	2.2	2.2	2.2	1.36	1.36	1.36
2	---	3.2	3.2	---	2.9	2.9	---	1.8	1.8
3	---	---	3.2	---	---	3.6	---	---	2.23
4	3.2	3.2	3.2	4.3	4.3	4.3	2.67	2.67	2.67
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
24	1.6	1.6	2.4	1.8	1.8	2	1.12	1.12	1.24

TABLE VIII
RELIABILITY INDICES FOR CIRCUITS FOR THE OPTIMAL SOLUTION

Circuit	SAIFI [interruptions]			SAIDI [hours]			ASAI [%]		
	Stage								
	1	2	3	1	2	3	1	2	3
1	3.2	3.2	3.2	2.39	2.41	2.57	99.97	99.97	99.97
2	0.8	0.8	0.8	1.60	1.25	1.25	99.98	99.99	99.99
3	0.8	1.2	1.6	1.60	1.70	1.66	99.98	99.98	99.98
4	0.4	0.4	0.8	0.80	0.80	0.99	99.99	99.99	99.99
5	1.6	1.6	2.4	1.80	1.63	1.92	99.98	99.98	99.98
6	---	0.8	0.8	---	1.60	1.20	---	99.98	99.99

1 during the three periods, and SAIFI is surpassed in circuit 5 during period 3.

In order to show the computation effort, the model solved has 14 932 variables and 14 725 equations, of which 7557 are discrete variables. The processing time is 12 h, 47 min on a PC with an Intel Core 2 Duo and 4-MB RAM.

VI. CONCLUSION

This paper presents an evaluation of a multistage optimization model for distribution systems expansion planning that takes into account: investment, maintenance, and operational (losses) costs, a three-level load representation, two investment alternatives for each feeder and substation, Kirchhoff voltages and currents laws, voltage drop constraints, and radiality constraints. The operational nonlinear cost is simplified by piecewise linearization and Kirchhoff voltages law by a linear disjunctive model. The resulting mixed-integer linear problem is solved by CPLEX. The model presents not only a single plan for a decision-maker but also a set of diversified plans with different costs, topologies, and features.

At the same time, the most relevant reliability indices are computed for each solution selected and, thereafter, the associated costs. The outcome shows that significant differences can occur in the cost incurred by a DISCO when the regulator applies penalties for not abiding by the limits for reliability indices and the estimated cost the customers are willing to pay for to receive redundancy in the system to avoid interruption. With this reliability information, it is possible to identify how individual nodes or consumers' sets are affected by every plan of the pool as well as the associated impacts on costs.

With the outputs of the model, it is possible to compare: 1) the cost impact to the utility when the regulator applies penalties for nonfulfillment of reliability indices and 2) the cost impact to customers for the damage caused by interruptions in the system.

Using this reliability information, it is possible to assess how an individual load or a customers' set is affected in every solution plan of the pool.

However, technical and economic analysis is not enough to enhance distribution efficiency. Furthermore, the network's optimality should be considered from various aspects, such as social and political factors, which need human decision making. Thus, there is no substitute for planners' judgment, regardless of how calculations or technical analyses are made.

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